

Creating Building Solutions

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by Neil Buckley

When Building Solutions, an Oakland, California, based company, joined the home performance industry in June 2006, we knew that recent advances in building science gave homeowners more options—and left them asking more questions—than ever before. With so many choices out there, whether among simple solutions like envelope sealing, or among big, expensive investments like PV solar arrays, homeowners needed to know which choice best suited their house and their budget. Our primary goal was to get the necessary information into the hands of homeowners so they could make educated decisions on their own, without feeling bullied into making improvements they didn't fully understand.

Of course, as anyone in the field knows, educating homeowners takes time. James Quazi, one of the founding partners at Building Solutions, had been working in the home performance industry for several years when he came to the conclusion that there had to be better, more efficient ways to get all the necessary information to the homeowner. That's when he decided to enlist me as a partner—someone who came from a technical background, creating software solutions for just this kind of problem. Together we developed a Web application to collect, process, and manage home performance data and present the results to the homeowner in a detailed, customized report.

We needed more than software, however, to build our own home performance contracting business. We

enlisted the help of a third partner, Matt Kealey, who has extensive experience making energy improvements, and who brought his own special brand of enthusiasm to the team. We began working with homeowners, testing their houses and making the necessary retrofits, all the while making improvements both to our software and to the ways we tested houses and responded to our clients. We started small, working one house at a time, air sealing attics and crawlspaces, installing furnace and duct systems, and replacing old water heaters. Our biggest challenge at that point was to strike a balance between finishing a job and finding more work. In many cases we would be working so hard to complete one job that upon completion we found ourselves scrambling because we had no new work lined up. But we kept at it, hired more employees, and expanded the scope of our work to include other improvements, such as radiant floor heating, and renewable energy sources, such as solar-thermal water heating. We found that our customers responded well to being offered a wide array of solutions, as opposed to feeling cajoled into purchasing one specific service.

Since our founding in 2006 until the time of this writing, we have air sealed and insulated 31 homes, installed 34 high efficiency HVAC systems with sealed duct systems, installed 6 radiant floor systems, replaced 23 water heaters with energy-efficient ones, and installed 12 solar-thermal systems. In our first year of business we did \$300,000 worth of business. In 2008, we expect that to grow to \$1,000,000.



CHRISTIAN FINK



MATT KEALEY

A solar thermal hot water system installed by Building Solutions provides domestic hot water and radiant floor heating.

As our process, and our software, improved, we became better organized and found it easier to get the work done while still taking the time to find more jobs. Establishing strategic partnerships helped immensely. We found that the easiest way to get more work was through referrals from contractors, architects, and homeowners who had worked with us before and knew firsthand our dedication to quality work. Most importantly, we made an effort to listen to customers, and make sure all their concerns were addressed. As Elizabeth De Oliveira, one of our satisfied customers put it, "It was a real pleasure to work with them. They not only know their stuff, but they were very accessible—always returning phone calls quickly, patiently answering all my questions. They were also willing to accommodate special requests, and I had several. I am very happy with the finished product."



Alicia Kealey

Matt Kealey of Building Solutions takes window measurements as part of a home performance test.

We also worked at improving our marketing presence, both through the Internet and through traditional media. We tried putting advertisements in magazines, the Yellow Pages, and other print materials, with limited success. We believe these types of marketing can be useful, as long as the investment is properly focused. We also invested some money with various internet search engines to increase our Web traffic, which over time has begun to pay off. Trade shows were another source of leads for us, and these were especially useful as they gave us a chance to talk to potential leads face to face right from the start. Marketing has been one of the toughest challenges for us, and we are still learning what works and what doesn't. In doing so, we have also learned the value of persistence

and patience; sometimes the best strategy is to make 1,000 pitches and see which one sticks.

Marketing Challenges

In many ways, the nature of the industry makes marketing the product particularly difficult. The basic principles of home performance contracting are simple enough that your average homeowner can grasp them rather easily. However, the field is just complicated enough that it still takes some one-on-one time with the homeowner to explain each home's potential problems, and the potential benefits of solving those problems, whether it be increased comfort, improved air quality, or reduced energy bills. Our software helps, but it can't replace the

need to work closely with our clients. And in some ways we wouldn't want it to. The most rewarding experience in our job is the moment when an idea clicks with a homeowner—when you can see in clients' eyes that they understand why it makes sense to do it a certain way. For example, one client was concerned that certain rooms in her home were too cold and other rooms were too hot. After taking air flow measurements from her registers and comparing those values with the load calculation requirements for each room, we were able to present her with a graph showing the drastic differences between the required air flow and the actual air flow; the data ended up confirming her complaints! She was immensely relieved to have her concerns understood and explained in a quantitative, scientific way. And knowing we had thoroughly investigated her problems, she felt very comfortable with the solutions we offered.

Of course, getting the chance to sit down with homeowners has been another major challenge. Homeowners want to improve their homes, but in most cases they don't even know where to start. We've found that many homeowners are immediately drawn to solutions like PV solar without understanding its place in the larger picture. Because PV has received intense media exposure lately, homeowners don't always know that there are more cost-effective solutions that they might want to explore before going solar. For example, in one particular project we had a couple who were extremely eager to install a PV system because of their skyrocketing energy bills. However, the system cost far more than they could afford to pay. Using our test, we were able to pinpoint a handful of areas where some simple, cost-effective fixes could make a huge difference. After we sealed the duct system to reduce AC expenditure, replaced an inefficient pool pump, and replaced all their incandescent lightbulbs with CFLs, their electricity bills were cut by over 40%. The couple could then purchase a PV system to handle the remainder of the electricity load at a price they could now afford.



ALICIA KEALEY

Members of the Building Solutions Team (from left to right): James Quazi, Neil Buckley, Scott Lowery, and Matt Kealey.



CHRISTIAN FINK

A blower door, one of the many tools used in a home performance test, helps measure air infiltration.

In this particular case, the homeowners were fortunate enough to be able to get their home tested and explore solutions. But in many cases, homeowners don't have this opportunity, simply because they don't know that there is a service like home performance contracting—a service that can help them find the most complete, most cost-effective way to improve their home. We believe that getting the word out is a challenge shared by all home performance contractors, and that they should meet this challenge in a spirit of mutual understanding and cooperation.

Finding Government and Utility Support

In spreading the word, home performance contracting faces another great challenge. The success of our industry depends heavily on the energy policies of the federal and state government, and of the public utility companies. If we want them to support our cause, we must reassure them by presenting them with the measured benefits of our work. And unlike the PV solar industry, for example, where benchmarking installations is as simple as comparing cost to watts, the home performance industry must deal with numerous variables—the scope of the project, the condition of the house before the project, weather

patterns, the client's lifestyle, and so on. For this reason, home performance contractors need to collect more data—years of utility data from before and after each installation—in order to impress those in state and federal government and utility leaders who can support us through tax breaks, tax credits, marketing, and so on.

Unfortunately, it is impossible to expect every home performance contractor, who must meet all the various demands of their day-to-day business, to take time to collect and report data, when there is no immediate reward. And most contractors are reluctant to keep returning to customers to get data for a job that was long since completed. For this reason, we believe that there should be solutions, preferably incorporating software, that make the process of collecting these data seamless and painless for the contractor. Ideally, the software that contractors use to create reports would have the ability to collect the necessary information and automatically report it to some central source. This is easier said than done, however, since it would also require utility companies to allow customer utility data to be collected electronically—something that they have been reluctant to do because customers argue that it would violate their privacy.

We believe that this hurdle could be overcome, by establishing a protocol allowing homeowners to give third parties access to their data for a specified window of time. Organizations such as one we belong to, the California Building Performance Contractors (CBPCA), are already working on improving the process of performance data collection, and with the help of other organizations and contractors, we'll come up with a solution.

Despite these challenges, we see great things ahead for our company and the home performance industry in general. As energy bills continue to increase, and homeowners becoming increasingly aware of the environmental impact of their energy usage, there will be more and more demand for our services. We look forward to meeting this demand in new and innovative ways—and in the process, to making homes healthier, more comfortable, and more energy efficient.

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